

Customer Churn Prediction and Segmentation Using Machine Learning Algorithms

Final Assignment



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Table of Contents

- Introduction and Market Analysis
- Exploratory Data Analysis
- Business Problem Definition
- Data Preparation and and Feature Engineering
- Experiments and Models
- Business Proposal
- References

Introduction and Market Analysis

Telecommunications is one of the most competitive industries, where customers can easily switch providers. Customer churn directly affects revenue, customer lifetime value (CLV), and acquisition costs, making retention a strategic priority.

The industry generates trillions of dollars globally and continues to grow through 5G, Artificial Intelligence (AI), Internet of Things (IoT), and Digital Services. At the same time, increasing competition has raised customer mobility and provider switching.

Key challenges include customer churn, market competition, price sensitivity, service quality issues, and rising customer expectations regarding support and personalization.

Machine Learning provides an opportunity to proactively identify customers at risk of churn, enabling targeted retention strategies, improved resource allocation, reduced revenue loss, and increased customer lifetime value.

Exploratory Data Analysis

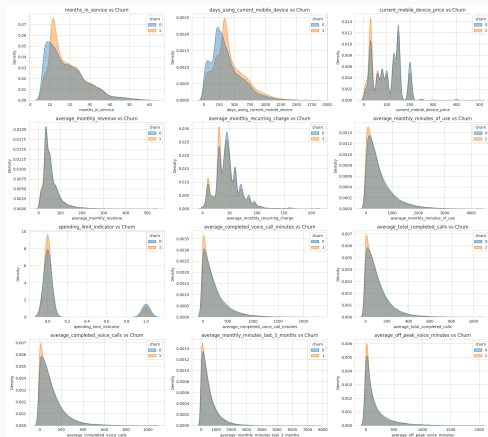


Figure 1. Density curves of the variables most correlated with churn.

Insight from Company A Data

The observed overlap between churn and non-churn distributions suggests that customer churn is not a linearly separable classification problem.

Business Problem Definition

Problem Context

Customer churn is one of the most significant challenges in the telecommunications industry, with reported churn rates reaching up to **20–30% annually** in highly competitive markets.

Retaining an existing customer is significantly less expensive than acquiring a new one, while customer churn directly impacts **revenue, Customer Lifetime Value (CLV), and profitability**.

Company A faces this same challenge. The dataset includes customer demographics, service usage metrics, revenue information, network quality indicators, device characteristics, and an explicit churn variable.

Business Objective

Reduce customer churn by identifying customers with a high probability of leaving the company and enabling proactive retention actions.

Business Problem Definition

Target Variable

Churn

- 0 = Remains
- 1 = Leaves

Primary Task: Churn Prediction

Models: Logistic Regression, XGBoost, CatBoost, LightGBM, and Ensemble Models.

Output: *Probability of churn.*

Secondary Task: Customer Segmentation

K-Means++ will identify groups of customers with similar behaviors.

Output: *Customer segments with different churn profiles.*

Data Preparation and Feature Engineering

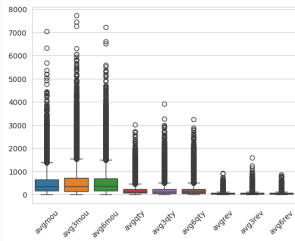


Figure 2. Distributions of customer usage and revenue variables.

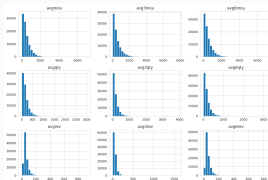


Figure 3. Boxplots of customer usage and revenue variables.

Data Preparation and Feature Engineering

Key Data Preparation Steps

- Missing numerical values were imputed using median-based strategies.
- Missing categorical values were imputed using the mode or an *Unknown* category when appropriate.
- Extreme outliers inconsistent with the variable context were identified and treated.
- Highly correlated variables and low-information features were removed.
- Variables with excessive missing values or a dominant *Unknown* category were excluded when appropriate.
- Categorical variables were encoded using One-Hot, Label, and Frequency Encoding according to their characteristics.
- Feature engineering was performed using the most important variables identified by the predictive models.

Experiments and Models

Table 1. Predictive Model Performance Comparison

Model	Dataset	Accuracy	Precision	Recall	F1 Score	ROC AUC
XGBoost	Train	0.6881	0.6822	0.6941	0.6881	0.7584
XGBoost	Test	0.6408	0.6354	0.6458	0.6405	0.6974
CatBoost	Train	0.6859	0.6786	0.6959	0.6871	0.7552
CatBoost	Test	0.6373	0.6311	0.6456	0.6382	0.6956
LightGBM	Train	0.6883	0.6822	0.6947	0.6884	0.7592
LightGBM	Test	0.6397	0.6348	0.6427	0.6387	0.6966
Voting Ensemble	Train	0.6893	0.6829	0.6967	0.6897	0.7593
Voting Ensemble	Test	0.6394	0.6334	0.6463	0.6398	0.6979
Logistic Regression	Train	0.6028	0.5984	0.6040	0.6012	0.6435
Logistic Regression	Test	0.6000	0.5953	0.6024	0.5988	0.6391

XGBoost achieved the best overall test performance (Accuracy = 0.6408, F1 Score = 0.6405, ROC AUC = 0.6974) and was selected as the final predictive model.

Experiments and Models

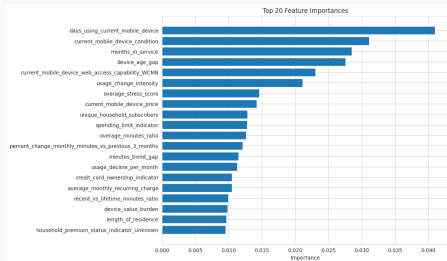


Figure 4. Top 20 feature importances of the XGBoost model.

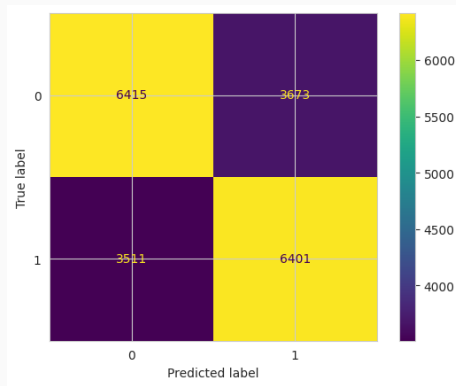


Figure 5. XGBoost confusion matrix on the test set.

Experiments and Models

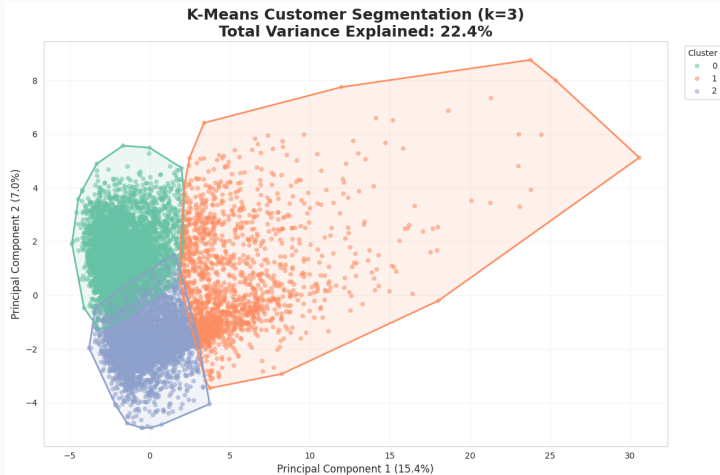


Figure 6. K-Means ++ customer segmentation ($k=3$) visualized using Principal Component Analysis (PCA).

Experiments and Models

Cluster 0 – Established Family Customers

Customers with strong family and residential stability, characterized by higher household income, homeownership, larger households, longer tenure, and high credit card ownership. They exhibit moderate service usage and a churn rate of **49.1%**.

Cluster 1 – High-Value Power Users

The most valuable customer segment, generating the highest revenue and service usage. They own more advanced devices, frequently upgrade them, and heavily use voice, data, and roaming services. Despite more service interactions, they show the lowest churn rate at **45.4%**.

Cluster 2 – Low-Commitment Individual Customers

Customers with weaker residential and family ties, lower homeownership, smaller households, and lower financial engagement. They present the highest churn rate at **51.5%**, making them the segment with the greatest attrition risk.

Business Proposal

Solution

Implement a Churn Prediction Retention Strategy that combines predictive modeling and customer segmentation to identify high-risk customers and apply targeted retention actions.

Business Impact

- Increase customer retention and Customer Lifetime Value (CLV).
- Reduce acquisition and marketing costs.
- Improve resource allocation through targeted campaigns.
- Preserve recurring revenue streams.

Expected Outcome

Even a small reduction in churn can generate significant financial benefits due to the large customer base and recurring nature of telecommunications services.

Telecommunications Industry

- GSMA – The Mobile Economy
- IBISWorld – Wireless Telecommunications Carriers in the United States
- EY – Telecommunications Industry Insights
- Verizon – Customer Mobile Plan Services

Customer Churn and Retention

- Predictive Analytics for Telecom Customer Churn
- Telecom Customer Churn Research (MDPI)

References

- Telecom Churn Prediction (ArXiv)
- Enhancing Customer Retention in the Telecom Industry with Machine Learning-Driven Churn Prediction

Business Resources

- Customer Retention Strategies in the Telecom Industry
- How to Reduce Churn Rate in the Telecom Industry

Thank You

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